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Research paper

Prediction of wind power ramp events via a self-attention based deep learning approach☆

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ABSTRACT

With the increasing popularity of wind power generation, the importance of accurate wind power ramp events (WPRE) prediction has been further emphasized. At present, WPRE prediction model based on deep learning shows superior performance, which does not need to fully grasp the operation of wind power generation, but ignores the interdependence mechanism of prediction error along the input characteristics of the neural network. This paper develops a self-attention (SAM) and wavelet transform (WT) based hybrid 1DCNN (one-dimensional convolutional neural network) and LSTM (long short term memory networks) for WPRE forecasting. In the proposed model, WT effectively separates the middle and high frequency noise signals in wind power and SAM can redistribute the neural weights in 1DCNN-LSTM, then 1DCNN-LSTM further extracts the space-time information of effective wind power. Finally, the performance of the forecast model proposed in this paper is tested by using the wind power data from the Belgian ELIA website. The experimental results show that, compared with the other six competition models, the proposed model effectively suppresses the influence of randomness, volatility and uncertainty of wind power on the recognition of ramp events. Furthermore, through quantitative analysis, this paper explains the reason of SAM is superior to traditional AM in terms of their application object and weight control range in the proposed model.

1. Introduction

Wind power has been widely exploited worldwide due to its advantages as renew-ability, pollution-free, and low cost. The total installed capacity of global wind power increased from 158 GW to 650 GW during 2009–2019 (ANON, 2019; Cui et al., 2021; Lin and Liu, 2020; Hache and Palle, 2019). The large fluctuation of wind speed in a short time period will cause wind power ramp events (WPRE) and pose a serious threat to the power quality and safe operation of the power grid (Gong et al., 2016; Duan et al., 2021; Aigner et al., 2012; Chang et al., 2018). Therefore, the importance of a precise WPRE prediction model is further emphasized by the increasing popularity of wind power penetration. If we can accurately predict the occurrence of WPRE in advance, various energy storage systems and fast hydro-power stations can be quickly employed as power reserves to reduce fluctuations in wind power output, thereby achieving real-time power supply demand balance and ensuring the safety and stability of the power grid (Cui et al., 2020; Xu et al., 2020).

A number of direct and indirect prediction methods have been developed for WPRE prediction (Zhang et al., 2018; Gallego-Castillo et al., 2015). The direct prediction method concentrate on the characterization of climbing events, such as climbing amplitude, duration and rate. For instance, Zareipour H et al. (2011) proposed a direct ramp event prediction method through data mining, a feature selection model based on mutual information was used to optimize the input of support vector machine (SVM) classifier to form a fine model for WPRE prediction. However, SVM is difficult to implement for large-scale training samples. Bossavy A et al. (2012) developed a probability model based on numerical weather prediction (NWP) to forecast WPRE, but the difference between the collected NWP information and the actual weather can cause considerable disruption to the forecast. Qu Y P et al. (2021) detected the climbing events in large historical wind power database through an adaptive algorithm, and established the statistical model of auto-correlation between multiple adjacent climbing events by data

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mining and association rule algorithm. Note that there is no universal way to make association rules. Therefore, the direct prediction is highly characterized and straightforward to implement, but which neglects wind power prediction and suffers from unavoidable forecasting deviations (Gallego-Castillo et al., 2015). By contrast, the indirect prediction method is based on the wind power prediction and predict WPRE according to the mechanism of wind power ramp events occurrence (Zhou et al., 2021).

Both of deterministic and probabilistic prediction methods have been commonly used in the indirect WPRE prediction method (Lange and Focken, 2006). Deterministic prediction is good at building mathematical relationship between meteorological, environmental information and the output of wind power by using factors such as atmospheric pressure, environment temperature, wind direction, wind farm location, topography, etc. However, deterministic prediction focus on the mathematical relationship between meteorological, environmental information and WPRE, these factors include atmospheric pressure, temperature, wind direction, wind farm location, topography and so on. Thus, deterministic prediction requires a large quantity of NWP information and computing resource, especially in short-term WPRE prediction (Chen et al., 2014). By comparison, probabilistic wind power prediction does not necessitate exogenous information in deterministic prediction, it focuses on the directly prediction of the probabilistic characteristics of the wind power. Therefore, it avoids the deterministic prediction error (Chen et al., 2010). Probabilistic prediction methods include integrated motion average model, auto-regressive model, Kalman filter Detector, Bayesian learning, machine learning model, etc (Koivisto et al., 2016; Hong et al., 2019; Wang et al., 2019). At present, machine learning methods (ML) attract more attention as their intelligent, and have shown excellent performance in the areas of image processing, natural language processing, speech recognition and vision-related tasks. It is worth mentioning that Khan, A. and other scholars (Khan et al., 2020) systematically reviewed the development process of convolutional neural networks (CNN) architecture. By discussing the remarkable progress CNN has made in image processing, the interest of researchers in machine learning is thus revived. Further, Khan, A. *et al (*Khan et al., 2021*)*.reviewed various applications of Genetic programming (GP) in image processing, providing useful resources for further research, and showing that as GP methods advance, its success in solving complex tasks, not only in image processing, but also in other fields, may increase. This will facilitate the exploration of machine learning applications in other fields. In view of the powerful nonlinear data fitting ability and parameter learning ability of ML-based methods, and with their successful application in speech recognition, image processing and other fields, the ML-based methods represented by artificial neural networks (ANN) have been gradually introduced into the wind power prediction field (Wan et al., 2020a; Cui et al., 2018, 2017; Chang et al., 2019).For instance, Cao Q et al. (2012) designed a recurrent neural network for wind power and WPRE prediction. Unfortunately, recurrent neural network cannot capture long range dependencies and are prone to disappearing gradients or exploding gradients. In order to solve the problem that a single model cannot fully learn, the fusion model based on machine learning has gradually become the mainstream development trend. For example, Zucatelli P.J et al. (2021) proposed a self-selected multi-layer coefficient correction model (CSMCC) by combining the back propagation (BP) neural network and the support vector machine (SVM) for WPRE prediction. Wan J et al. (2020b) developed a hybrid WPRE prediction model to combine CNN and long short term memory networks (LSTM). Han L et al. (2020) used deep neural networks (DNN) and improved multi task learning (MTL) architecture to accomplish WPRE prediction. Although the above fusion models show excellent predictive performance in the field of WPRE, the fusion methods are too simple, and there is still a large room for improvement. In order to enhance the performance of the prediction algorithm with scanty samples, Qureshi et al. (2017) tried modeling by employing transfer-learning to transfer information from the old wind farms to new one, and experiment results demonstrate the validity. A.V. Zhukov et al. (2016) proposed a feature mining method for WPRE prediction and used the existing benchmark test method to verify the competition and robust performance within its hourly prediction range for the method. Compare with the deterministic methods, neural network models improved the WPRE prediction accuracy. However, they neglect the interdependence mechanism of forecast errors among input features of the neural networks in WPRE prediction, and reduces the robust capability of the neural networks under the influence the randomness, volatility and uncertainty of the wind power. In 2014, the google mind team proposed the attention mechanism (AM) to improve the feature extraction capabilities of neural network models, furthermore, AM has been applied in image processing and machine translation and the neural networks gained the capability of weight assignment for the input features (Chen et al., 2021, 2020; Li et al., 2020; Lang et al., 2021; Lu et al., 2020). However, AM directly acts upon the input features in most cases, it is not sufficient to cope with the inherent volatility and randomness of the wind power in WPRE prediction (Zhu et al., 2019; Tang et al., 2020; Song et al., 2020).

Additionally, wind power have the nature of high randomness and rapid fluctuation, and noise is one of the main factors, thus it is difficult to forecast the original time series directly. Therefore, the researchers integrated the commonly used sequence decomposition techniques into the fusion model to remove noise in wind power and improve data quality, mainly including, empirical mode decomposition (EMD) (Dm et al., 2020; Wu et al., 2020), ensemble EMD (EEMD) (Santhosh et al., 2018), variational mode decomposition (VMD) (Duan et al., 2021; Wu et al., 2020; Han et al., 2022) and so on. For EEMD and EMD, the most serious problem is mode mixing, which not only causes serious aliasing of time-frequency distribution, but also causes a single intrinsic mode function to lose its clear physical meaning. VMD can solve the mode mixing phenomenon existing in EEMD and EMD, but its performance depends on the selection of the total number of modes and there is no general method. It can only be selected by referring to relevant experience or using specific search algorithm. If not selected properly, it will still lead to the mode mixing of each component. Wavelet transform (WT) has the idea of local transform, which provides a time-frequency window that changes with frequency. It is an ideal tool for signal time-frequency analysis and processing. Wavelet transform can decompose wind power signal into low frequency part and high frequency part, and the noise is mostly contained in the high frequency part. Therefore, the noise contained in the high frequency part is filtered out specifically, and then the low frequency part and the high frequency part which have been de-noised are reconstructed, namely, the inverse wavelet transform, so as to achieve the purpose of wavelet signal denoising. In fact, the variation of the high frequency is a manifestation of the randomness of wind power, that is, the excessive subdivision of the high frequency band has little physical significance. Therefore, compared with the excessive signal decomposition of EMD, EEMD and VMD, WT can not only effectively remove high-frequency noise, but also require less computation and have higher cost performance.

To bridge this gap in WPRE prediction, this paper proposed a self-attention mechanism (SAM) (Cao et al., 2020; Xu et al., 2021)and WT based hybrid one-dimensional convolutional neural network (1D-CNN) and LSTM (WT-1DCNN-LSTM-SAM) for probabilistic forecasting of WPRE, which reconstructs input features, reduces the dimension of feature vectors, and completes the extraction of wind power features, in addition, SAM redistribute the neural weights of 1DCNN-LSTM networks and update the hidden layer output of LSTM. Furthermore, this method does not need to consider the impact of the input on the weights, and directly retains the important characteristics of the short-term wind power sequence in the neurons. Therefore, efficiency of attention distribution and convergence of the neural network are significantly improved by SAM. One wind power dataset from the Belgian ELIA website is used to verify the superior performance of the proposed WT-1DCNN-LSTM-SAM model for WPRE prediction in 4 seasonal patterns. Furthermore, the significant effectiveness of the proposed WT-1DCNN-LSTM-SAM model is validated via systematic comparisons with the state-of-the-art models.

Main contributions of this paper are summarized as below:

1) A novel fusion neural network model is developed for WPRE prediction, which can effectively reduce the influence of randomness, volatility characteristics in wind power on the recognition of ramp events.

2) The notion of the self-attention mechanism is introduced to fusion neural networks for wind power ramp forecasting. SAM can directly reset the weight distribution for different hidden layer output sequences of LSTM by the end-to-end training, and it solves the problem of neural network information overload and improves the adaptive ability of the artificial neural networks to the uncertainty of input features.

3) Mechanisms of AM and SAM are determined by comparative study of the AM and SAM based neural network models in WPRE prediction. Processes of weight distributions in these two models are analyzed and superiority of SAM is further confirmed through the training mechanisms.

2. Data processing of wind power ramp events

2.1. Definition of wind power ramp events

WPRE mainly refer to one-way and large fluctuation of wind power in a short time caused by long-term extreme weather, which includes climbing events and downhill events. When strong low-pressure system (or cyclone), low-level jet, thunderstorm, thunder shower or similar long-term extreme weather events occur, the power of wind farm will increase sharply and climbing events of wind power will appear. In this regard, WPRE easily causes frequency out of limit or instability, which has an adverse impact on the power grid.

To construct accurate WPRE, the characteristic parameters of WPRE are defined as Fig. 1 (Gong et al., 2016). The red curve in Fig. 1, which is defined as the increasing direction of wind power, it can be seen as a climbing event. On the contrary, the blue curve is described as a downhill event. Under the initial conditions of t_0 and t_1, where ΔP_u and ΔP_d are incremental and decremental quantity of wind power during Δt_u and Δt_d, respectively. Therefore, the indirect definition of WPRE could be shown as:

$$|P_{t+\Delta t} - P_t| > P_{th} \quad (1)$$

Where P_t is the wind power at the time of t and P_{th} is the threshold value of the wind power. According to (1), if the absolute value of the power difference exceeds the given wind power threshold in a fixed period, WPRE may occur. Therefore, accurate prediction of wind power in a certain range is the premise to effectively identify the climbing events.

2.2. Introduction and processing of the experimental data

To evaluate the performance of the proposed WT-1DCNN-LSTM-SAM, one realistic case study was conducted using the historical data of wind power, which were collected from ELIA Group website, from January 1st, 2019 to November 30st, 2020. ELIA Group operates the offshore and onshore wind farms stations located 30 km away from Zeebrugge port in Belgium and is authorized for the operating energy market. Fig. 2 shows that wind power generation constitutes approximately 16 % of installed generation capacity in Belgium, which could increase the credibility of the experiment to a certain extent. Significantly, according to relevant planning, by the end of 2021, the total installed capacity of wind power generation will reach 6 GW, which can supply about half of the country's electricity (ELIA).

In this paper, wind power datasets with 15-minute resolution covering four seasons in 2019 are trained for the purpose of capturing seasonal atmospheric patterns, while wind power datasets in 2020 are tested for verifying the prediction performance of different models. In view of this, the Δt presented in (1) is set as 15-minute, thus 96 sampling data points are obtained the whole day. Finally, the P_t presented in (1) is set as 3 % of total wind power installed capacity by considering the experience of a large number of researchers and the characteristics of Belgian wind power, which provides a crucial basis for predicting wind power climbing events in this paper (Han et al., 2020; Jing et al., 2021).

The box-plot in Fig. 3 shows the distribution of training datasets and test datasets in the same quarter. Among them, the upper and lower limits of the box are respectively the upper and lower quartiles of the data, which means that the box contains 50 % of the data, and its size reflects the fluctuation degree of the data to some extent. In other words, the flatter the box, the more centralized the data distribution. The line below the box represents the minimum value, and conversely the top line represents the maximum value. In addition, the line in the middle of the box represents the median line, which represents the average level of data. Further, a probability density plot of the data distribution has been added to Fig. 3 to further understand the boxplot and reflect the actual distribution of the data. For each quarter, the distribution of training data has been aligned with test data. As these results elucidate, suppose

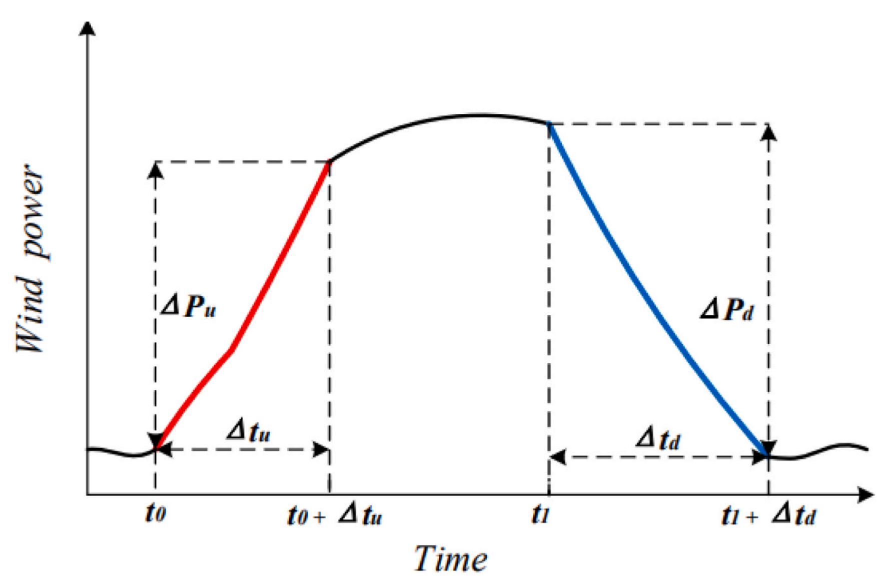

Fig. 1. Characterization of Wind Power events.

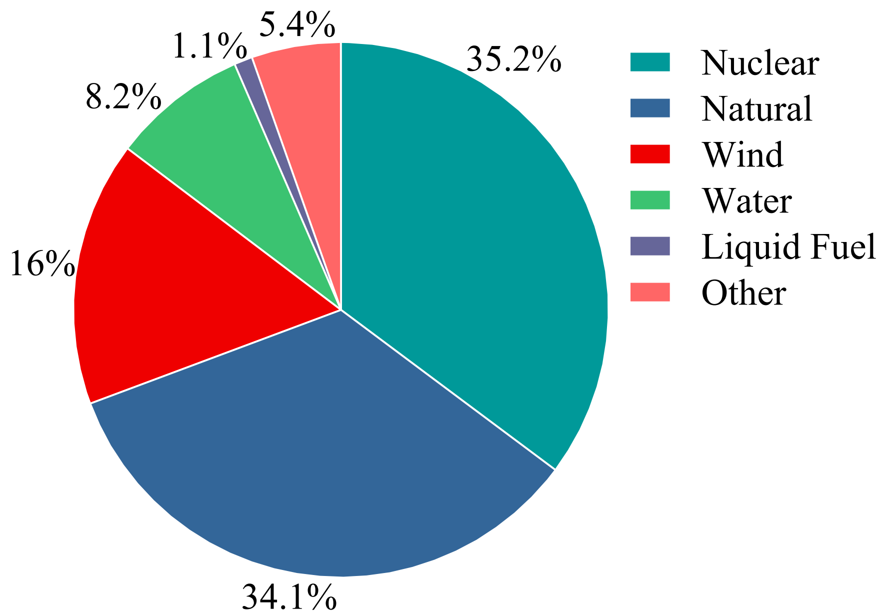

Fig. 2. Proportional distribution of installed power generation capacity in Belgium.

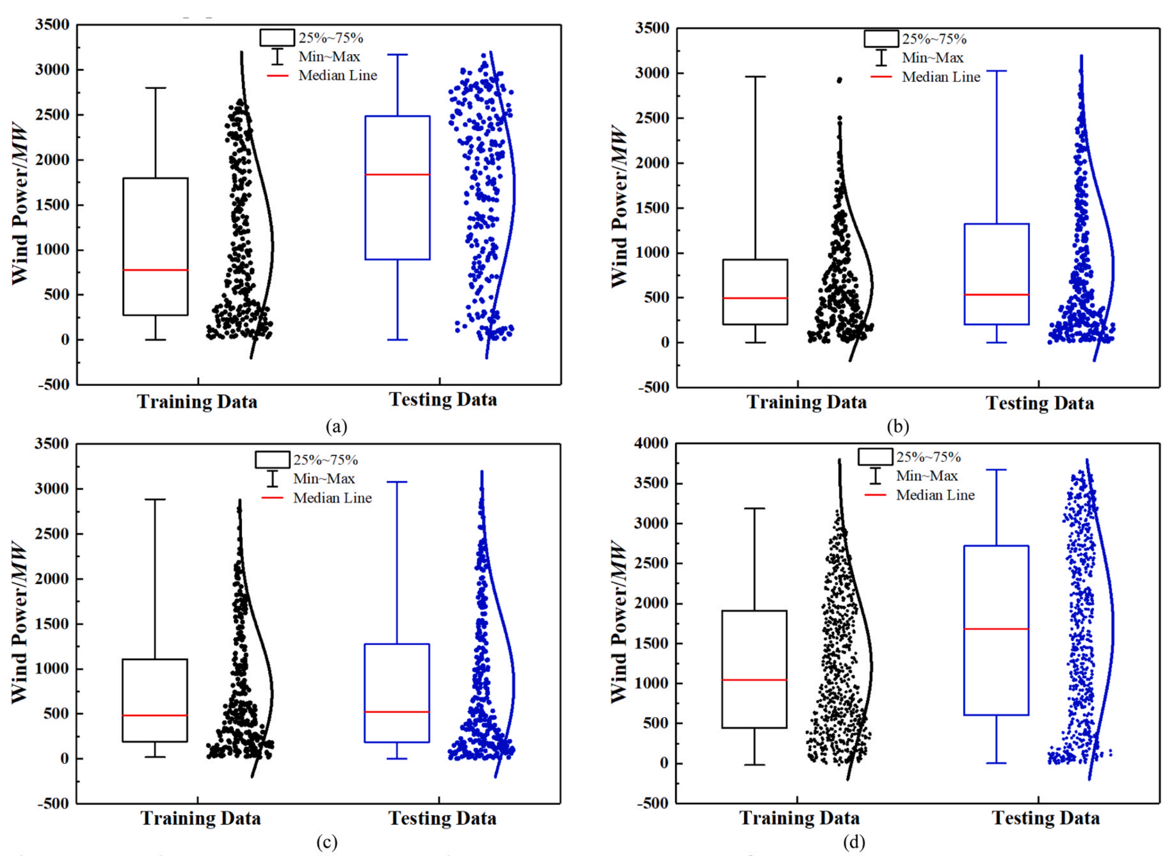

Fig. 3. Distribution of testing data and training data in four quarters: (a) First quarter; (b) Second quarter; (c) Third quarter; (d) Fourth quarter.

the strategy of quarterly independent modeling could be adopted, based on the outstanding advantages of above discussed, a more accurate WPRE prediction model will be obtained to employ as the prediction engine without loss of generality. Furthermore, those of first and fourth quarter appear to be more uniform and their Gaussian curves are more closer to the standard normal distribution, which shows that the wind power data of the above quarter have certain regularity and predictability to the training of the model with more conducive, while others are negatively skewed hint at the non-Gaussian properties. Since the distribution of negative skewness, it is not conducive to the prediction of the points in the wind power curve range with large wind power value, thus affecting the generalization ability of the models. Therefore, a stronger generalization ability is required for the proposed probabilistic forecasting model in the second and third quarters. It is worth noting that the four quarters have large differences in box size, which indicates that the fluctuation degree of the four quarters' data is not consistent, reflecting the importance of the quarterly breakdown. Due to the seasonal distributional diversity, the proposed WT-1DCNN-LSTM-SAM model is trained separately for different quarters. Moreover, the method of quarterly independent modeling above conclude the proof.

2.3. Wavelet transform

High-quality wind power generation data can address several severe impacts of the inherent volatility and randomness behavior of wind power on power system balance in the electricity market. Traditionally, due to the existence of measurement error, noise and other interference information in the wind power data, which will lead into dispensable errors. In this regard, in order to establish a more accurate prediction model, it is necessary to obtain a few practical and effective methods for preprocessing the wind power data.

On the basis of the above, WT is employed as the denoising method for the wind power without loss of generality, which fully utilizes its own prominent advantages to adjust the sampling step size in time domain for different frequencies (Pjz et al., 2021).

In this paper, in order to better fully utilize the wavelet denoising prominent advantages, this paper focuses on the coie5 as wavelet basis function , meanwhile, a 7-level wavelet decomposition layer is presented to acquire a higher degree data information based on a large number of trial-and-error experiments. Fig. 4 shows the comparison of wind power generation of both training and test datasets using wavelet denoising in four quarters, which provides insightful information about the efficacy of the proposed approach in denoising. It is found that the wind power generation possesses a high level of volatility and intermittency, and therefore the proposed wavelet denoising method is employed to solve the inherent inferiority. In addition, the high frequency noise curve accords with the characteristics of Gaussian white noise distribution, which can be seen as an approximate standard noise, proving that WT can achieve the expected effect and the rationality of the selected parameters. Notably, as the Fig. 4 elucidates, compared with the original wind power curves, the curve (low frequency signal) denoised by WT possesses superior performance in each quarters, which shows more stronger stability and retains the important characteristics, such as useful signal spike and mutation signal.

In summary, WT fully excavates the effective information of data, reduces the influence of random fluctuation factors, and improves the accuracy and stability of prediction. In contrast, the traditional method of moving smoothing filtering will not only remove the high-frequency noise, but also lose the useful high-frequency signal to a certain extent.

2.4. Selection of time window

In order to comprehensively analyze the dependence of time series, an experiment based on time windows chosen appropriately is proposed. Due to the higher dependency between time series, it is necessary to select an appropriate time windows to capture these dependencies precisely.

Firstly, the wavelet denoised datasets are sampled by sliding window method, and the input samples with the size of D@S is generated, which are used to train and test the fused model. Where S and D are defined as time windows width and data feature dimension respectively.

Secondly, suppose that the first input sample sequence is set as $S_1=[x_1 , x_2..., x_S]$, after sliding L steps in the sliding time window, the second input sample sequence is $S_2=[x_{1+L} , x_{2+L}..., x_{S+L}]$, where $x_i \in R^{1\times D}$. Therefore, if there are φ sampling points in the training datasets, the total (φ-S×L+1) sample sequences can be obtained.

Finally, based on the sample sequences above, a label is added to each sample sequence correspondingly.Suppose the predicted length of time step is N (N≥1), then the label of S_1 is expressed as $y1=[x_{S+1} , x_{S+2}..., x_{S+N}]$, as well as the label of S_2 can be expressed as $y2=[x_{S+L+1} , x_{S+L+2}...x_{S+L+N}]$. By this way, corresponding labels will be obtained of all sample sequences without loss of generality.

Indispensably, sensitivity analysis was carried out to determine the size of time window width and sliding step. As an instance, if the value of window width is larger, the correlation between sample sequences will become sparser, thus the computational burden is increased, and the inherent characteristics of extracting wind power will not be conducive. By calculation, the window width(S) and the sliding step is set as 7 and 1 (Equivalent to a 15-min sampling interval) in this paper (Cui et al., 2021), respectively, these two parameters can better meet the relevant experimental requirements. In other words, when modeling each quarter separately, the historical data of the first 1.75 hours (7 data points) is selected as the characteristics to predict the wind power value of the next time node (15 min), the single-step prediction is realized, which can minimize the prediction error of wind power. Meanwhile, the specific selection process of window width can be shown in Section 3.3.

3. Summary of the related models

3.1. Summary of AM and SAM

In this subsection, this paper introduces the attention mechanism briefly and describe the way of applying self-attention into the proposed fusion model.

Traditionally, a standard neural network can be regarded as a non-linear transformation from input to fixed-dimensional potential

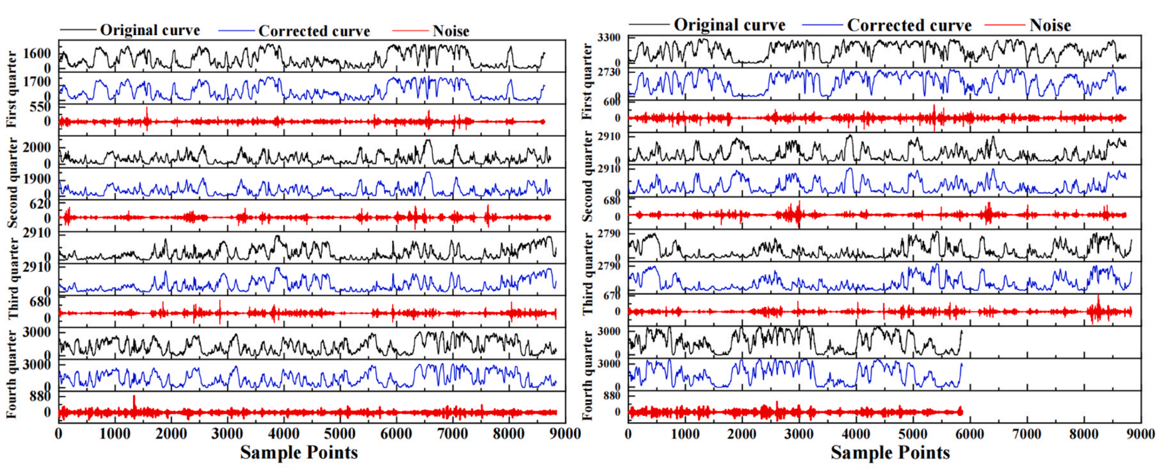

Fig. 4. Data denoising by wavelet transform: (a) Data wavelet denoising in training data; (b) Data wavelet denoising in testing data.

representation. This paradigm causes difficulties to control the information flow in neural network and affects the final result. For a typical example in machine translation under a popular encoder-decoder architecture (Song et al., 2020), the entire sentence as input is encoded into a fixed-length of hidden representations via a recurrent neural network (RNN) or CNN. When generating the output of target translated sentence, the network utilizes the same context or hidden representations at every step. This leads to a bottleneck for these analogical sequence-to-sequence tasks.

AM is used to solve this problems and it aims to address the main defects of conventional neural networks so as to obtain the capability of weight assignment for the input features and reduce the difficulty of prediction. It is naturally that the feature with larger weights plays a more important role in the output of the proposed model. As a side benefit, AM improves the feature extraction capabilities of neural network models, and relieves the problem of neural network information overload to a certain extent.

Due to the outstanding characteristics of structuring training and test samples discussed before, when the sliding time window is applied to wind power data, the sample features obtained have stronger correlation. However, AM is not completely effective for the prediction of wind power and its ramp events concurrently under these circumstances, which will be verified in the next part of the case analysis.

To overcome the shortcomings of the AM in the prediction of wind power and its ramp events, the stronger correlation between the output of the hidden layer of the LSTM and the output of the network is took into account in this paper. Crucially, due to the certain sparsity of the output of LSTM hidden layer, the outputs of the network have different degrees of influence under the action of different hidden layer units (Duan et al., 2021; Kuja and Bck, 2021). If only a few outputs of the hidden layer units is assigned a larger weight, the problem of information overload in the neural network will be solved. From the above discussion, to acquire such a crucial bond, this paper addresses a new method based on adding self-attention layer between network output layer and LSTM output layer, which improves the adaptive ability of the artificial neural networks to the uncertainty of input features and transcends such a crucial barrier of traditional attention mechanism. Therefore, the approximation ability of the proposed model will be further improved.

Fig. 5 shows the self-attention structure proposed in this paper and h_1 is set as an example to indicate the calculation process of SAM. h=[h_1 , h_2…,h_n] is the output of LSTM hidden layer unit. When a query vector q_1=[q_1,q_2…,q_n] and a key vector k_j=[k_1,k_2…,k_n] are set respectively, thus, the similarity score between q_1 and each k_j can be calculated by the scaled-dot-product function shown in (2), where q_i, k_i and v_i are obtained by (6) respectively. d represents the output dimension of LSTM layer. Subsequently, the similarity scores (expressed as) is normalized in the range of [0,1] by the softmax function, as illustrated in (3). Moreover, in order to obtain the corresponding weight of h_1, α_1 should be calculated by (4). Finally, the parameter $\widehat{h}_1$, which is updated by h_1 weighting, can be expressed as (5) and is set as the input of the lower network.

SAM is characterized by differentiable calculations, and allows the end-to-end training in the neural networks. As shown in (6), the appropriately learnable weight matrixes of W_Q, W_K and W_V, are obtained ceaselessly through the end-to-end training. Then the rest of output values of LSTM hidden layer are updated by significantly different weight assignments, according to (2)-(6).

$$\alpha_{1,j} = \frac{q_1 k_j}{\sqrt{d}} (j = 1, 2, 3\cdots\cdots n) \tag{2}$$

$$\widetilde{\alpha}_{1,j} = \frac{e^{\alpha_{1,j}}}{\sum_{j=1}^{n} e^{\alpha_{1,j}}} (j = 1, 2, 3\cdots\cdots n) \tag{3}$$

$$\alpha_1 = \sum_{j=1}^{n} \widetilde{\alpha}_{1,j} v_j (j = 1, 2, 3\cdots\cdots n) \tag{4}$$

$$\widehat{\mathrm{h}}_1 = h_1 \alpha_1 \tag{5}$$

$$\begin{cases} q_j = h_j W_Q \\ k_j = h_j W_K (j = 1, 2, 3\cdots\cdots n) \\ v_j = h_j W_V \end{cases} \tag{6}$$

3.2. Summary of our proposed forecasting method

Based on the outstanding characteristics of WT, 1D-CNN, LSTM and SAM, the proposed prediction model is implemented in a cascaded WT-1DCNN-LSTM-SAM learning framework shown in Fig. 6. The 1D-CNN takes every subsequence stemming from WT as inputs, aiming to extract the implicit features of tight coupling relationship between wind power data. Subsequently, LSTM takes the outputs of flattening layer in CNN as inputs to further extract the temporal correlation hidden features. Finally, SAM redistributes the neural weights of 1DCNN-LSTM networks and updates the hidden layer output of LSTM. The modeling and implemented process of WT-1DCNN-LSTM-SAM is described

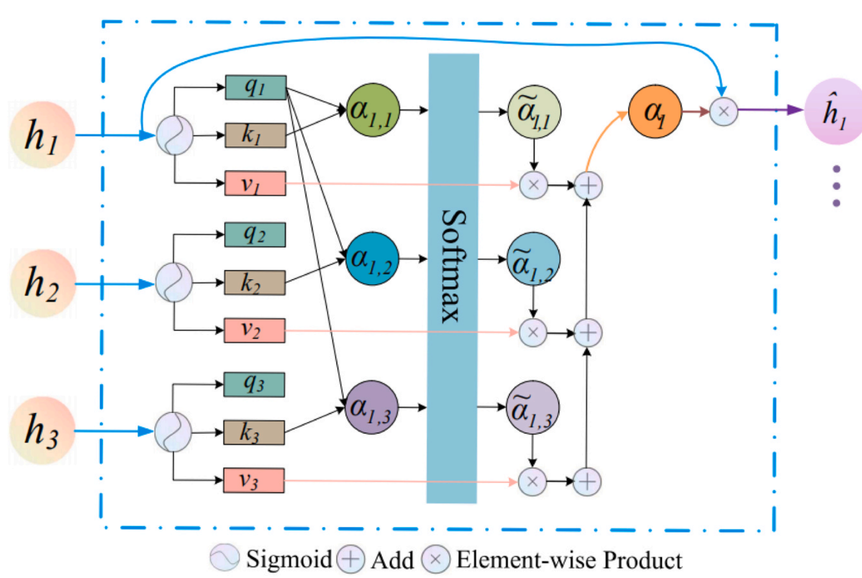

Fig. 5. The structure of SAM.

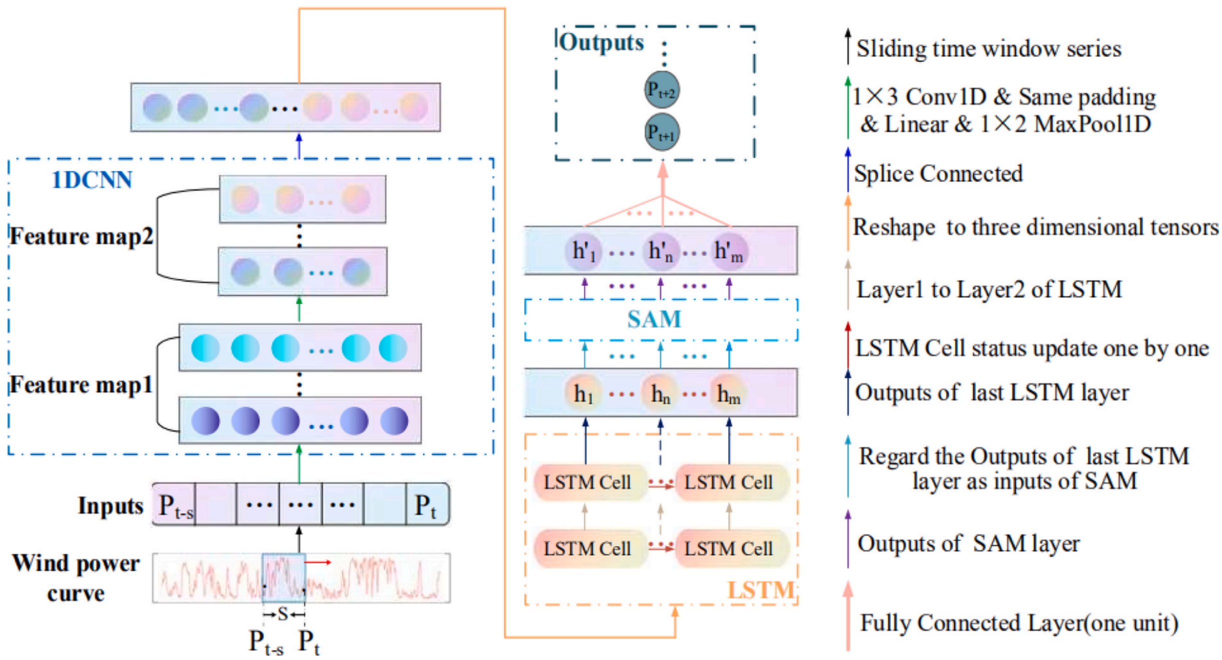

Fig. 6. The structure of our proposed model based on 1DCLSTM-SAM.

succinctly as Algorithm 1.

Algorithm 1. : The modeling process of proposed **WTCLSA** Approach

1 **Start:**Load wind power data and Divide by quarter
2 **Data Preprocessing:** Wavelet denoising, Data normalization, Constructing training samples by sliding time windows and divided into training set and verification set.
3 **Initialization:**Learning_rate=0.001, Batch_size=64, Dropout rate=0.5, Epoch=300, LossFunction=MAE, Activation function=Linear
4 **Start Model Training:**
Input: Training set and Verification set
for i = 1,2, …, Epoch **do:**
5 Use the training set, through forward propagation and back propagation, an iterative process is completed
6 Use the validation set to calculate the validation loss value
7 **If** validation loss value is decreasing**:**
8 Update the model parameter matrix and save
9 **Else:**Return step **4**, go on training model
10 **End for**
Output: Trained model for WP and WPRE prediction
End Model Training
End

Therefore, the proposed fusion model obtains significant effectiveness and higher accuracy for the short-term prediction of wind power and its ramp events, which is driven by the end-to-end network training and continuous parameter optimization of the above process.

3.3. Selection of network parameters

Fig. 6 shows the structure of our proposed model. It can been seen from Fig. 6 that 1D-CNN adopts the structure of 4-layer network, with alternating one-dimensional convolutional layer (Conv1D) and one-dimensional pooling layer (MaxPool1D). In the 1D-CNN part of this model, the size of convolution kernel and pooling layer were set as 1×3 and 1×2 respectively, the number of the convolution kernel was set as 80, and the length of the convolution step and pooling step size all were set as 1.

Meanwhile, LSTM adopted a three-layer hidden layer structure, and the number of cyclic cores in the hidden layer was set as 200. To address such a crucial barrier called over-fitting, the dropout layer had been added after each LSTM hidden layer and the random drop probability was set as 0.5. Finally, part of SAM in this algorithm needs to be designed independently and has a similar structure to the traditional neural network hidden layer, therefore, the number of neurons was set as 200 based on a large number of trial-and-error experiments. At the

Table 1
Parameter settings of the model.

Type	Parameters
First layer of Conv1D	80(size=1×3)
Second layer of Conv1D	80(size=1×3)
First layer of LSTM	200
Second layer of LSTM	200
Third layer of LSTM	200
SAM	200
Dense	1
convolution step	1
pooling step	1
learning_rate	0.001
batch_size	64
epoch	300
Loss Function	MAE
Sliding Step	1
Window Width	7
Optimizer	Adam
Dropout	0.5
Pooling Size	1×2
Activation function of CNN and LSTM Layer	*linear*
Activation function of SAM Layer	*tanh*

same time, it can also avoid the over-fitting phenomenon in the network training process and reduce the complexity of the network.Subsequently, the output of the updated LSTM hidden layer were taken as the input of the fully connected layer with one neuron, and continuous output predictive values could be obtained. Note that the hidden layer of both CNN and LSTM is activated by *linear* activation functions, while the hidden layer of SAM is activated by *tanh* activation functions. The loss function of the proposed model is set as the mean absolute error (MAE). Adam algorithm is used to optimize the parameters in the back propagation of network training (Liu et al., 2021a; Shahid et al., 2021). Compared with the traditional gradient descent algorithm, Adam combines the advantages of momentum algorithm and RMSprop algorithm, and uses the exponential weighted average of gradient to update the weight w and deviation b, thus reducing the longitudinal fluctuation in the optimization process, improving the optimization speed and training efficiency. Moreover, to ensure the rapidity and convergence of training, Learning_rate, Batch_size and Epoch were set as 0.001, 64, and 300 respectively, then effectiveness of the above parameters setting related to the network model will be verified in the case analysis subsection. Detailed Settings of model parameters are shown in Table 1.

Not to be ignored, the selection of width of the time window is the key step to realize the prediction of WPRE. If the selected time dimension is too long, the contribution of the distant historical data to the current prediction data will be small, making the training of the model unable to reach the effect, thus affecting the prediction accuracy of the model. On the contrary, if the selected time dimension is too short, some data features will be lost and key information may be lost, resulting in low prediction accuracy of the model. Based on the wind power data of the fourth quarter after wavelet denoising, Fig. 7 shows the changes of MAE value and MAPE value of WTCLSA prediction model under different widths of the time window. It is found that when the width of the time window is 7, the WTCLSA prediction model has the minimum MAE value and MAPE value in all four quarters. Therefore, when the width of the time window is set as 7 in this paper, it is more conducive to the recognition of WPRE.

4. Evaluation criteria

4.1. Forecasting metrics of wind power

In this paper, RMSE, MAE and MAPE were used to reflect the degree of dispersion of prediction results, the error between the predicted value and the real value, and the average level of prediction errors respectively (Shahid et al., 2021; Yin et al., 2021; Zn et al.). MAE and RMSE are used to assess the predictive ability and accuracy of the proposed prediction method, while MAPE is used to evaluate the relative deviation of point forecasting results from real wind power outputs. Meanwhile, R^2 is used to evaluate the goodness-of-fit between the predicted value and the real value. The definitions of these metrics are presented in (7)-(10) respectively. Where m describes the total number of samples, h_i and$\widehat{h}_i$ represent the predictive results and the real data of wind power at the *i-th* time step respectively, $\overline{h}_i$represent the mean value of original data at the time step t_i.

$$RMSE = \sqrt{\frac{1}{m}(\sum_{i=1}^{m}|h_i - \widehat{h}_i|)} \tag{7}$$

$$MAE = \frac{1}{m}(\sum_{i=1}^{m}|h_i - \widehat{h}_i|) \tag{8}$$

Table 2
Forecast Result of Wind Power Ramp Events.

		Observation	
		True	False
Prediction False	True	TP	FP
	False	FN	TN

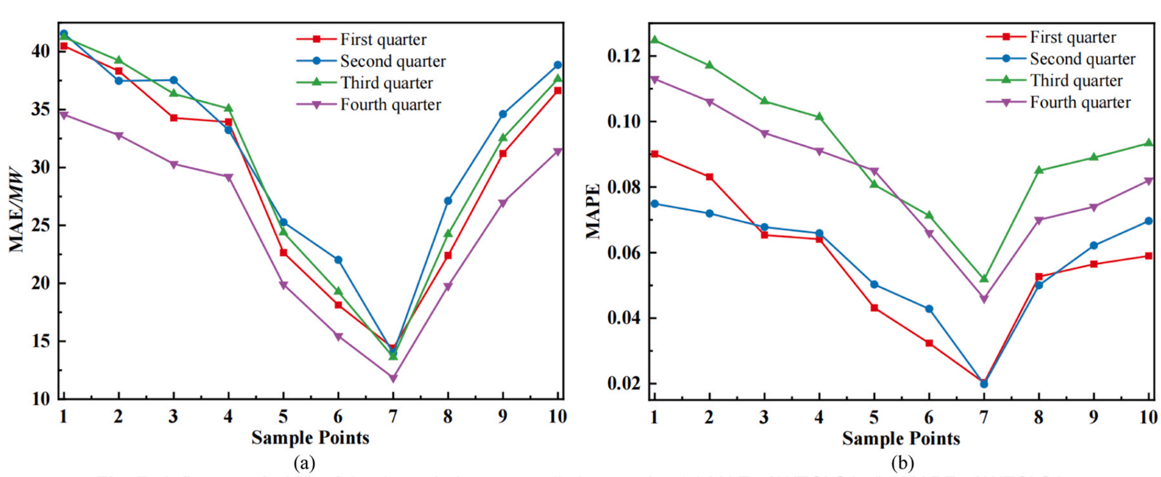

Fig. 7. Influence of width of the time window on prediction results: (a) MAE of WTCLSA; (b) MAPE of WTCLSA.

$$MAPE = \frac{1}{m}\sum_{i=1}^{m}\left|\frac{h_i - \widehat{h}_i}{h_i}\right| \times 100\% \tag{9}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{m}(h_i - \widehat{h}_i)^2}{\sum_{i=1}^{m}(\overline{h}_i - \widehat{h}_i)^2} \tag{10}$$

4.2. Forecasting metrics of WPRE

As shown in Table 2, there are 4 kinds of results in WPRE prediction. TP and TN indicate that the prediction results are consistent with the actual results, while TN means that the prediction and actual results have not occurred. The meanings implied by FP and FN are opposite to the above discussion. Furthermore, FP and FN are called false positive (Misinformation) and false negative (Underreporting) respectively in this paper.

According to the above discussion, Fa (Correct rate), Rc (Recall rate), Ur (Underreporting rate), Ep (Misinformation), and Er (Misjudging probability of climbing direction) are chosen as evaluation metrics for WPRE performance, which should be satisfied according to the numerical relationship of (16). The definitions of these metrics are presented in (11)-(15) respectively. Subsequently, to more accurately reflect the prediction ability of the model and avoid the impact of TN value on the statistical results of TP, TN and FP (TN is relatively large), TN value is not considered in this study. In addition, Du is set as the number of wrong predictions of climbing direction in TP, as shown in (12) and (15).

$$R_c = \frac{TP}{TP+FN} \times 100\% \tag{11}$$

$$F_a = \frac{TP-DU}{TP+FP+FN} \times 100\% \tag{12}$$

$$U_r = \frac{FN}{TP+FP+FN} \times 100\% \tag{13}$$

$$E_p = \frac{FP}{TP+FP+FN} \times 100\% \tag{14}$$

$$E_r = \frac{DU}{TP+FP+FN} \times 100\% \tag{15}$$

$$F_a + U_r + E_p + E_r = 1 \tag{16}$$

5. Example analysis and comparison

5.1. Experiments and analysis

For validating the effectiveness and superiority of the proposed WT-1DCNN-LSTM-SAM (WTCLSA) model, six neural networks forecasting methods are used as benchmarks, including multi-layer perceptron (MLP) 、CNN、RNN、LSTM、1DCNN-LSTM (CL) and AM-1DCNN-LSTM (ACL) (Liu et al., 2021a; Shahid et al., 2021; Yin et al., 2021; Zn et al.; Kuja and Bck, 2021). It should be noted that all models are trained and tested based on the data denoised by wavelet transform. Additionally, in order to ensure comparability, the number of layers and the number of convolution kernels designed for each reference model is controlled at a similar level to that of the proposed model

Indirect WPRE prediction method is adopted in this paper, thus wind prediction should be carried out firstly. Some wind prediction results are

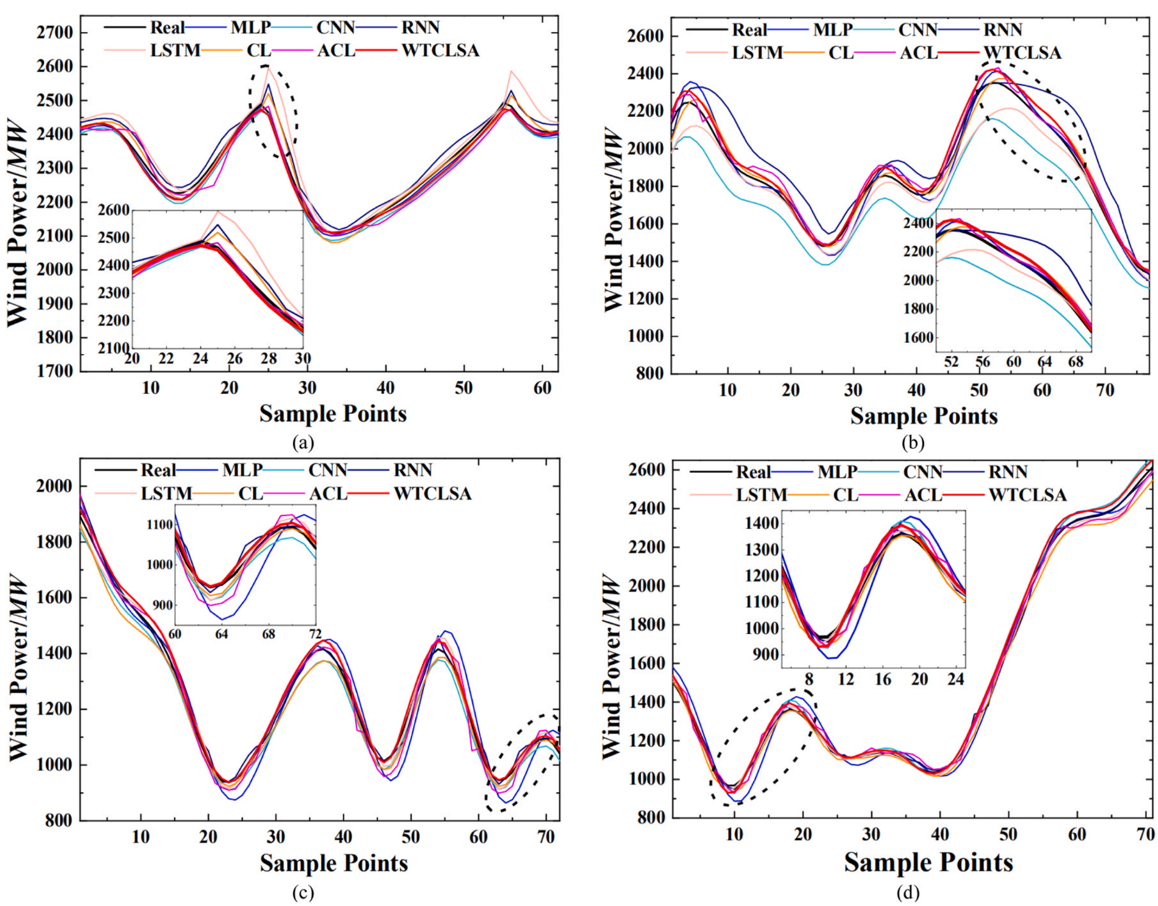

Fig. 8. The part samples wind power forecast results of different algorithms in wavelet denoising conditions: (a) First quarter; (b) Second quarter; (c) Third quarter; (d) Fourth quarter.

Table 3
Wind power forecasting metrics of different algorithms in wavelet denoising conditions.

Quarter	*Index*	MLP	CNN	RNN	LSTM	CL	ACL	WTCLSA
First	**MAE**	23.7732	28.9327	15.2688	24.5633	15.0992	23.0534	**14.4801**
	RMSE	34.2273	34.1855	21.8345	33.9347	20.4951	32.2681	**18.2831**
	MAPE	4.37 %	5.26 %	2.88 %	4.46 %	3.83 %	4.27 %	**2.03 %**
	R^2	0.99817	0.99817	0.99925	0.99820	0.99934	0.99837	**0.99948**
Second	**MAE**	15.5126	50.7125	29.2071	19.1968	15.1693	18.5731	**13.9674**
	RMSE	23.4733	70.8653	56.4089	38.6954	24.0304	24.7565	**20.1702**
	MAPE	2.29 %	14.97 %	4.08 %	3.96 %	3.43 %	3.63 %	**1.98 %**
	R^2	0.99880	0.98780	0.99307	0.99674	0.99874	0.99867	**0.99911**
Third	**MAE**	24.4701	23.5842	20.3263	17.9918	18.9482	18.2560	**13.6217**
	RMSE	30.5547	31.9504	29.6090	33.9872	28.2177	24.5425	**18.0545**
	MAPE	7.92 %	18.97 %	11.28 %	9.45 %	15.76 %	14.45 %	**5.18 %**
	R^2	0.99807	0.99789	0.99819	0.99761	0.99835	0.99875	**0.99932**
Fourth	**MAE**	26.2875	29.2070	19.5922	19.8528	27.7102	21.0804	**11.8426**
	RMSE	37.3878	36.5832	27.1546	32.4289	36.9412	27.3714	**15.8789**
	MAPE	7.34 %	12.24 %	8.81 %	10.50 %	11.31 %	9.18 %	**4.60 %**
	R^2	0.99842	0.99848	0.99916	0.99881	0.99846	0.99915	**0.99971**

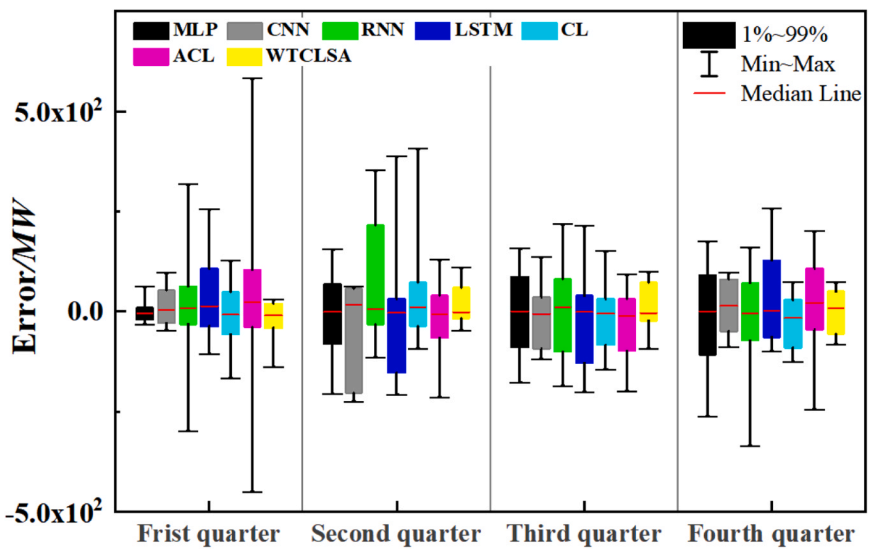

Fig. 9. Box plot of error distribution of each algorithm in wavelet denoising conditions.

randomly selected from the 4 seasons in the test set, and they are illustrated in Fig. 8. Fig. 8(a)-(d) shows the prediction results of spring, summer, autumn and winter, respectively. Inset figure shows enlarged details in Fig. 8. As seen from Fig. 8, the wind power shows the characteristics of randomness and volatility in 4 seasons. The prediction accuracy and stability of each model are reflected by the following behaviors of the actual wind power fluctuations. According to Fig. 8, WT-1DCNN-LSTM-SAM (WTCLSA) shows better following ability than other 6 models.

Table 3 shows the wind power forecasting metrics of different algorithms in wavelet denoising conditions. As shown in Table 3, the proposed algorithm obtains the best prediction results in four seasons. The average MPAE of the proposed algorithm in four seasons is 3.45 %, which is 56.22 % lower than that of ACL (7.88 %). The range of MAE in WTCLSA is between 11.84 and 14.48, and the average of MAE is 13.48mw, which is 33.46 % lower than that of ACL. In addition, compared with MLP, CNN, RNN, LSTM and CL, the average MAPE of WTCLSA is reduced by 37.04 %, 73.17 %, 48.96 %, 51.68 % and 59.79 % respectively.

Fig. 9 shows the error box plots of different algorithms in 4 seasons under wavelet denoising conditions. As seen from Fig. 9 that, compared with the other six models, the error distribution range of WTCLSA is smaller and closer to 0. This means that WTCLSA has better accuracy and stability in wind power prediction.

Fig. 10 shows some WPRE prediction results, and the WPRE forecasting metrics of different algorithms in wavelet denoising conditions are listed in Table 4. In Fig. 10 (a)-(d), incorrect recognition of climbing direction is not appearing in WPRE prediction, which indicates that WT can effectively separate the high-frequency noise signal of the original wind power signal and reduce the influence of random fluctuation factors of wind power.

Table 4 shows the WPRE forecasting metrics of different algorithms in wavelet denoising conditions. As seen in Table 4. The average Fa of WTCLSA is 91.94 %, which is 59.67 %, 37.05 %, 12.73 %, 93.79 %, 49.33 %, and 26.89 % higher than that of ACL, MLP, CNN, RNN, LSTM, and CL, respectively. In addition, compared with other 6 models, Rc, Ep, and Ur of WTCLSA has been significantly improved. Notably, as seen in Fig. 8(b) and (c), forecast curve is much lower than the real curve in the second and third quarter, thus, Ep of CNN in the second and third season of Table III is less than that of WTCLSA, but it does not mean that the prediction performance of CNN is better than WTCLSA, because this phenomenon may be also caused by an increasing of Ur in CNN.

5.2. Analysis of WT

WT is used to replace the traditional denoising method in WPRF prediction, thus the effect of WT on prediction results is demonstrated this section. Fig. 11 shows the prediction metrics of wind power. In this

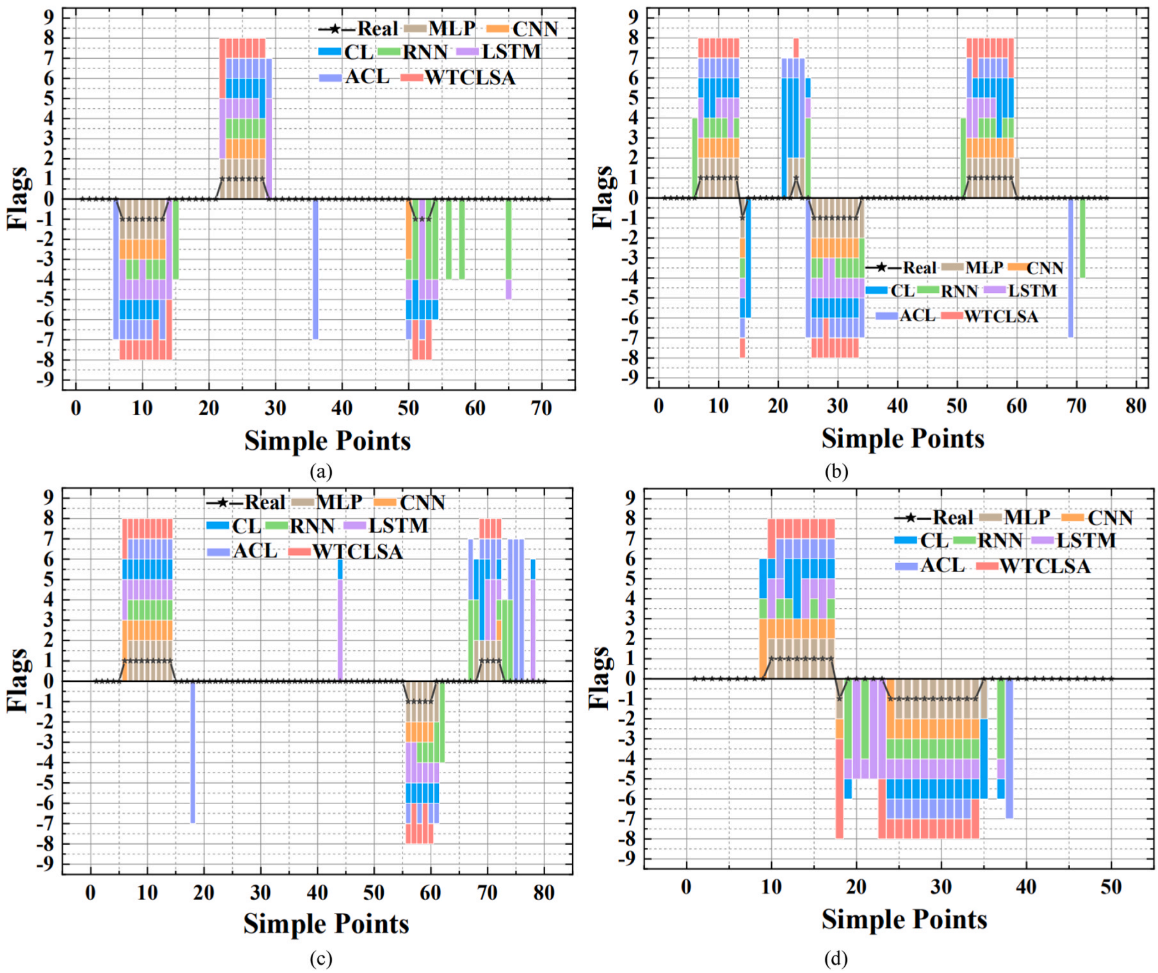

Fig. 10. The WPRE forecast results of different algorithms in wavelet denoising conditions: (a) First quarter; (b) Second quarter; (c) Third quarter; (d) Fourth quarter.

Table 4
The WPRE forecasting metrics of different algorithms in wavelet denoising conditions.

Quarter	*Index*	MLP	CNN	RNN	LSTM	CL	ACL	**WTCLSA**
First	***Rc***	94.68	87.77	82.45	82.98	88.30	84.57	**96.28**
	Fa	66.92	81.68	51.16	55.52	69.46	55.48	**92.35**
	Ep	29.32	6.93	37.95	33.10	21.33	33.57	**4.08**
	Ur	3.76	11.39	10.89	11.38	9.21	10.25	**3.57**
	Er	0.00	0.00	0.00	0.00	0.00	0.71	**0.00**
Second	***Rc***	96.21	76.52	76.52	88.64	90.08	89.39	**99.24**
	Fa	70.17	75.37	46.12	60.94	79.17	59.60	**93.58**
	Ep	27.07	**1.50**	39.73	31.25	13.73	33.33	**5.71**
	Ur	2.76	23.13	14.15	7.81	7.10	7.07	**0.71**
	Er	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
Third	***Rc***	93.92	87.29	72.93	80.66	85.08	88.95	**97.79**
	Fa	63.43	84.49	44.59	65.18	69.37	54.76	**92.19**
	Ep	32.46	**3.21**	38.85	19.20	18.47	38.44	**5.73**
	Ur	4.11	12.30	16.56	15.62	12.16	6.80	**2.08**
	Er	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
Fourth	***Rc***	93.07	95.67	74.03	87.88	88.31	82.25	**97.40**
	Fa	67.82	84.67	47.90	64.63	71.83	60.51	**89.64**
	Ep	27.13	11.49	35.29	25.72	18.66	26.43	**7.97**
	Ur	5.05	3.84	16.81	9.33	9.51	13.06	**2.39**
	Er	0.00	0.00	0.00	0.32	0.00	0.00	**0.00**

figure, subscript 1 represents the case where WT is absent and subscript 2 represents the case where WT is used. It can be concluded that, WT effectively separates the high-frequency noise signal of the original wind power signal, reduces the influence of random fluctuation in wind power, then the important characteristics of the original signal is retained, which plays an important role in the prediction of wind power and its climbing events.

5.3. Analysis of SAM and AM

According to the prediction results of WPRE in *5.1*, WTCLSA performs better in almost all indicators. Notice that although AM is

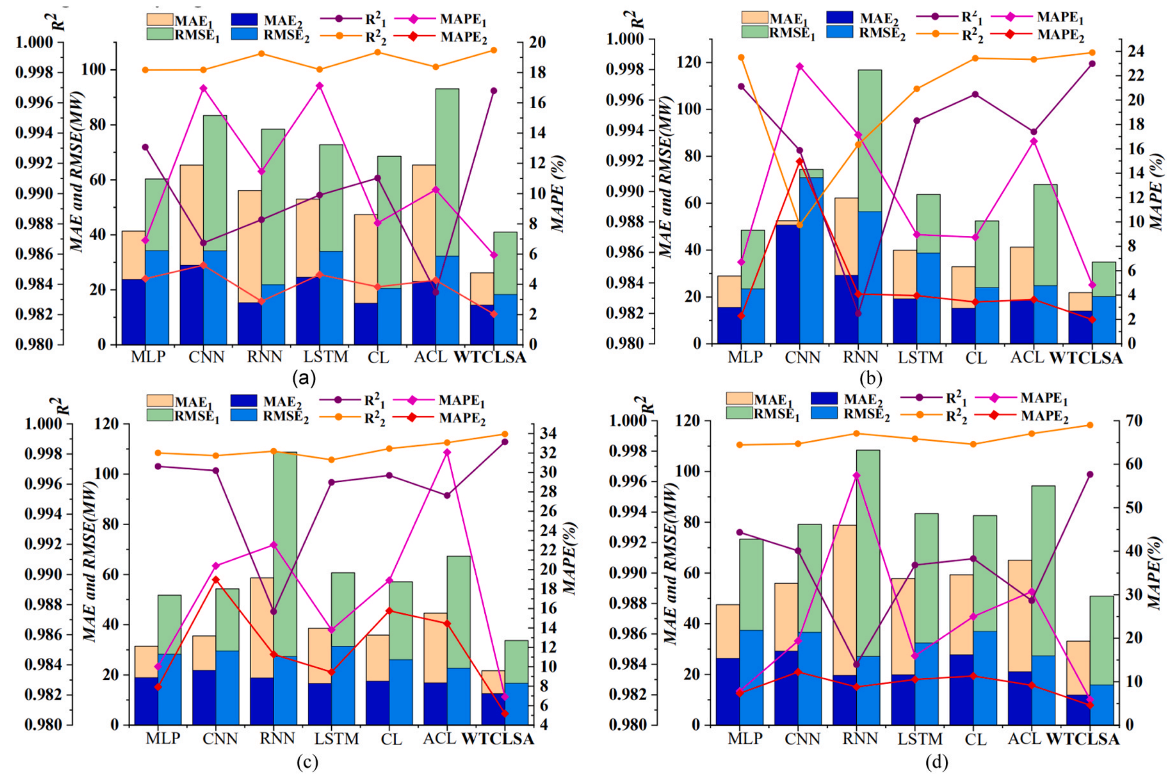

Fig. 11. Comparison on the wind power prediction metrics: (a) First quarter; (b) Second quarter; (c) Third quarter; (d) Fourth quarter.

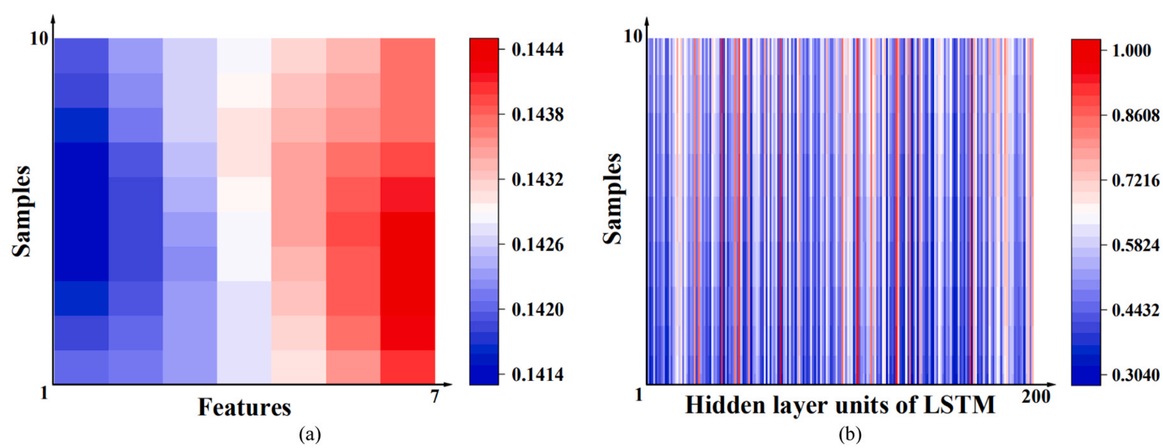

Fig. 12. The part samples weight distribution of AM and SAM: (a) AM; (b) SAM.

Table 5
Weight control of AM and SAM in test samples.

Methods	Application object	Weight control range (Fourth quarters)	Range of all seasons
AM	Eigenvalues of wind power test samples	[0.13005,0.15727] [0.14189,0.14386] [0.13001,0.15854] [0.13001,0.15339]	[0.13001,0.15854]
SAM	Output values of hidden layer units of LSTM	[0.23815,1.0000] [0.50633,1.0000] [0.24949,1.0000] [0.17830,1.0000]	[0.17830,1.0000]

employed, ACL does not show better prediction performance. Therefore, different mechanisms of AM and SAM in WPRE prediction is studied in this section. Theoretically, AM can solve the problem of information overload in WPRE forecasting. When an input feature has a greater impact on the output of the network, the corresponding input feature will be assigned a larger weight by AM, then the feature of this input features is amplified, in such a way that reduces the impact of random volatility on WPRE forecasting. However, as can be seen from Table 4, Fig. 8 and Fig. 10, compared with CL, the prediction performance of ACL is not improved. The reasons are studied as follows, the number of sample eigenvalues of wind power is not large (determined by the designed time window), while WT reduces the random fluctuation of wind power and preserves the important features of the original signal, which strengthens the correlation between sample features. Therefore, AM considers that the importance of each eigenvalue to the network output is nearly the same during the test process. Thus, the weight distribution is approximately the same, and as a result, AM does not effectively amplify the role of important input features.

In order to verify the above analysis, the weight distribution diagram of sample eigenvalues during the train process is shown as Fig. 12 (a) (This sample is randomly selected in the data set). In Fig. 12 (a), the abscissa is the index number of the input eigenvalue, and the ordinate represents the index number of some training samples. As can be seen from Fig. 12 (a), in the process of model training, only 7 input variables can be adjusted. Futhermore, in the prediction process of WPRE, the detailed weight control information of AM and SAM in test samples is shown is Table 5. Notice that and the weight adjustable range of AM is only [0.13001,0.15854] (The whole adjustable range of the weight is [0,1]). By comparison, Fig. 12(b) shows the weight distribution of SAM during the train process. The abscissa in Fig. 12 (b) is the index number of some training samples, and the ordinate corresponds to the index number of LSTM hidden layer unit. As seen from Fig. 12 (b) and table IV, the number of the adjustable features of SAM is 200, and the adjustable range of weight is [0.17830, 1.0000], therefore, the adjustable features and range of SAM are significantly higher than that of AM.

Furthermore, the action mechanism of SAM are studied as follows. When the model is trained end-to-end, the output of the LSTM hidden unit and the real output label are connected through the density layer, thus there is a strong correlation between LSTM and the output label. Therefore, the gradient value of the loss function has a direct relationship with the output of the LSTM hidden unit in the process of back propagation. Therefore, according to the importance of hidden layer units to network output and the similarity between different hidden layer units, SAM assigns the corresponding weights to each hidden layer unit, and multiplies the assigned weights with the corresponding historical output of LSTM hidden layer units, then a new LSTM hidden layer unit output is obtained. In this way, SAM can directly reset the weight distribution for different hidden layer output sequences of LSTM, which greatly expands the range of weight adjustment since the number of the hidden layer units is larger than the input features. Therefore, SAM not only solves the problem of information overload of neural network, but also enhances the adaptive ability of artificial neural networks to the uncertainty of input features.

5.4. Analysis of computational efficiency

Computational efficiency for model construction is critical to the development practicality of the wind power forecasting methods. Without loss of generality, the model training time in fourth quarter is reported in Table 6. It can be seen that, the time consumption of WTCLSA is close to that of CNN, RNN and ACL, but much less than that of LSTM under the same physical hardware environment. Therefore, it is worth noting that although WTCLSA increases the network depth, the training time does not increase significantly due to the improvement of its computational efficiency.

5.5. Comparison with the existing research

To verify the advanced nature of the proposed method, Refs (Zhang et al., 2021; Liu et al., 2021b, 2021c; Wang et al., 2021; Cheng et al., 2021). are selected for comparison research, due to the input features of the prediction models in these references are similar to our proposed model. These reference models mainly include CNN-LSTM prediction model based on spatiotemporal correlation characteristic matrix, which is denoted as MSC, deep and transfer learning framework(DETL), CNN with tailored inputs(TICNN), augmented convolutional network (ACNN), hybrid models based on the K-shape and K-means guided Convolutional Neural Network integrating Gated Recurrent Units (KK-CNN-GRU). The relative prediction error MAPE in Refs (Zhang et al., 2021; Liu et al., 2021b, 2021c; Wang et al., 2021; Cheng et al., 2021). has been directly used in Table 7. At the same time, in order to make a comprehensive comparison with the existing models, the representative competition models in Refs (Zhang et al., 2021; Liu et al., 2021b, 2021c; Wang et al., 2021; Cheng et al., 2021). are also used as the comparison models in this paper. In order to ensure the fairness of the comparison, MAPE, which reflects the overall prediction accuracy, is selected as the evaluation index. The detailed comparison results are shown in Table 7. It can be seen that compared with the best forecast model ACNN in Refs (Zhang et al., 2021; Liu et al., 2021b, 2021c; Wang et al., 2021; Cheng et al., 2021)., MAPE of the proposed model WTCLSA is reduced by 26.3 %, which shows that the method proposed in this paper is advanced.

Table 6
Model training time of different forecasting methods.

Methods	MLP	CNN	RNN	LSTM	CL	ACL	**WTCLSA**
Time(s)	247	367	320	907	261	393	**348**

Table 7
The MAPE of these comparison methods.

Methods	MAPE
MSC (Zhang et al., 2021)	0.0956
SS (Zhang et al., 2021)	0.5026
DETL (Liu et al., 2021b)	0.1377
IRDNN (Liu et al., 2021b)	0.1455
IAEDNN (Liu et al., 2021b)	0.1422
UDETL (Liu et al., 2021b)	0.1851
TICNN (Wang et al., 2021)	0.1833
FR-CNN (Wang et al., 2021)	0.2201
DC-CNN (Wang et al., 2021)	0.2319
ARIMA-kal (Wang et al., 2021)	0.2504
ACNN (Cheng et al., 2021)	**0.0467**
SVR (Cheng et al., 2021)	0.2268
KK-CNN-GRU (Liu et al., 2021c)	0.1910
MLR (Liu et al., 2021c)	0.2015
FCDNN (Liu et al., 2021c)	0.2080
CNN-GRU (Liu et al., 2021c)	0.2175
GRU-FC (Liu et al., 2021c)	0.1975
PM (Liu et al., 2021c)	0.2010
ARIMA (Liu et al., 2021c)	0.2135
AR(1) (Liu et al., 2021c)	0.2145
VMD-ILSTM (Han et al., 2019)	0.0807
CNN-LSTM (Han et al., 2020)	0.0803
STSR-LSTM (Ahmad and Zhang, 2022)	**0.0665**
LMBNN (Ahmad and Zhang, 2022)	0.0885
BRBNN (Ahmad and Zhang, 2022)	0.0885
Combined based on VMD-EMD (Wu et al., 2020)	0.1757
Combined based on VMD-EWT (Wu et al., 2020)	0.1730
VMD-BIS-ILSTM (Han et al., 2022)	1.0326
WTCLSA	**0.0344**

More importantly, to further strengthen the validation of our proposed model, VMD-ISTM in Ref (Han et al., 2019)., CNN-LSTM in Ref (Han et al., 2020)., LMBNN, BRBNN and STAR-LSTM in Ref (Ahmad and Zhang, 2022)., Combined based on VMD-EMD and Combined based on VMD-EWT in Ref (Wu et al., 2020)., and VMD-BIS-ILSTM in Ref (Han et al., 2022). are also cited as comparison methods. All of the above are advanced hybrid forecasting methods, consistent with the dataset used in this paper sourced from the ELIA website. The detailed comparison results are also shown in Table 7. It can be seen from Table 7 that compared with the best forecast model in Refs (Han et al., 2020, 2019; Ahmad and Zhang, 2022; Wu et al., 2020)., MAPE of the proposed model WTCLSA is reduced by 48.3 %, showing the best predictive performance. Notably, Some comparison models include time sequence decomposition techniques such as VMD, EMD and EWT, which can also demonstrate the effectiveness of WT in this paper to a certain extent.

In short, through comprehensive comparison and analysis with existing studies, it is shown that the model proposed in this paper has high superiority, high prediction accuracy, and can improve the accuracy of WPRE prediction.

6. Conclusion

WPRE prediction is crucial for dealing with the uncertainty of power grid operation and ensuring the safe and stable operation of wind power grid. At present, WPRE prediction models based on deep learning does not need to fully grasp the operation of wind power generation, but they ignore the interdependence mechanism of forecast errors along input features of the neural networks. This paper develops a SAM and WT based hybrid 1DCNN and LSTM for probabilistic forecasting of WPRE, benefited from the redistribution of the neural weights in 1DCNN-LSTM networks by the proposed SAM and the separation of the high frequency noise signal in wind power by WT, WTCLSA effectively suppresses the influence of randomness, volatility and uncertainty of wind power on the recognition of ramp events. Furthermore, this paper compares the different mechanisms of traditional AM and SAM in WPRE prediction, and proves the superiority of SAM by quantitatively analyzing their scope of action in weight allocation. Six neural network forecasting methods and models of open publications are used as benchmarks, and numerical studies based on actual wind farm with four seasons validate the accuracy and stability of the proposed WTCLSA model. In general, the proposed methodology is promising to support the online WPRE prediction and ensure a safe and stable operation of wind power grids.

Author statement

We would like to submit the enclosed manuscript entitled “Prediction of Wind Power Ramp Events via a Self-Attention Based Deep Learning Approach”, which we wish to be considered for publication in journal of “Energy Reports”. No conflict of interest exits in the submission of this manuscript, and manuscript is approved by all authors for publication. I would like to declare on behalf of my co-authors that the work described was original research that has not been published previously, and not under consideration for publication elsewhere, in whole or in part. All the authors listed have approved the manuscript that is enclosed.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The authors do not have permission to share data.

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